

PLamb — Consolidated Results Update

(Pantheon+SH0ES, V10V6c)

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Executive summary

SNe-only fits continue to give reduced $\chi^2 \approx 1.166$ across all models. Model selection prefers Λ CDM once parameter penalties are applied (BIC strongest). FR(STF-only) and KdS(no- Λ) are effectively identical in fit quality.

Allowing a variable- c with Λ improves χ^2 by < 1 for one extra parameter; AIC/BIC still prefer Λ CDM.

Λ CDM — parameters & metrics

Parameter	Value
H_0	71.87
Ω_m	0.216
Ω_Λ	0.373
χ^2	1980.15
Dof	1698
χ^2/dof	1.166
AIC	1986.2
BIC	2002.5
N	1701
acceptance	0.646

KdS ($\Omega\Lambda=0$) — parameters & metrics

Parameter	Value
H0	50.31
Ω_m	0.396
$\Omega\Lambda$ (fixed)	0
A_acc	1.038
n_acc	0.659
w_eff (median -/+)	-0.810 -0.121/+0.123
χ^2	1979.86
dof	1697
χ^2/dof	1.167
AIC	1987.9
BIC	2009.6
N	1701
acceptance	0.423

FR (STF-only; $\Omega_\Lambda=0$, $\gamma_c=\epsilon_M=0$) — parameters & metrics

Parameter	Value
H0	50.40
Ω_m	0.398
Ω_Λ (fixed)	0
A_acc	1.030
n_acc	0.654
w_eff (median -/+)	-0.815 -0.117/+0.124
χ^2	1979.87
dof	1697
χ^2/dof	1.167
AIC	1987.9
BIC	2009.6
N	1701
acceptance	0.417

FR (variable-c + Λ ; A_acc=n_acc= ϵ _M=0) — parameters & metrics

Parameter	Value
H0	71.87
Ω_m	0.006
Ω_Λ	0.441
γ_c	-0.183
χ^2	1979.37
dof	1697
χ^2/dof	1.166
AIC	1987.4
BIC	2009.1
N	1701
acceptance	0.542

Model deltas (A minus B)

Λ CDM – KdS

Parameter	Value
$\Delta\chi^2$	0.29
ΔAIC	-1.71
ΔBIC	-7.15
Δdof	1

Λ CDM – FR(STF)

Parameter	Value
$\Delta\chi^2$	0.28
ΔAIC	-1.72
ΔBIC	-7.16
Δdof	1

Λ CDM – FR(var-c+ Λ)

Parameter	Value
$\Delta\chi^2$	0.79
ΔAIC	-1.21
ΔBIC	-6.65
Δdof	1

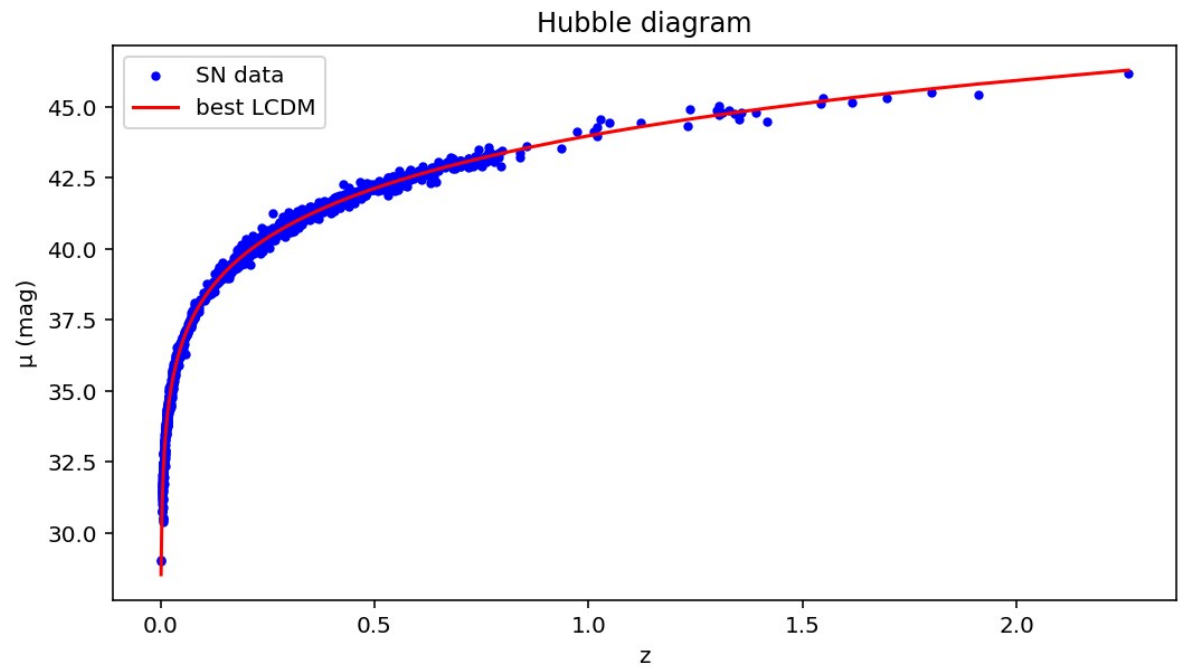
FR(STF) – KdS

Parameter	Value
$\Delta\chi^2$	0.01
ΔAIC	0.01
ΔBIC	0.01
Δdof	0

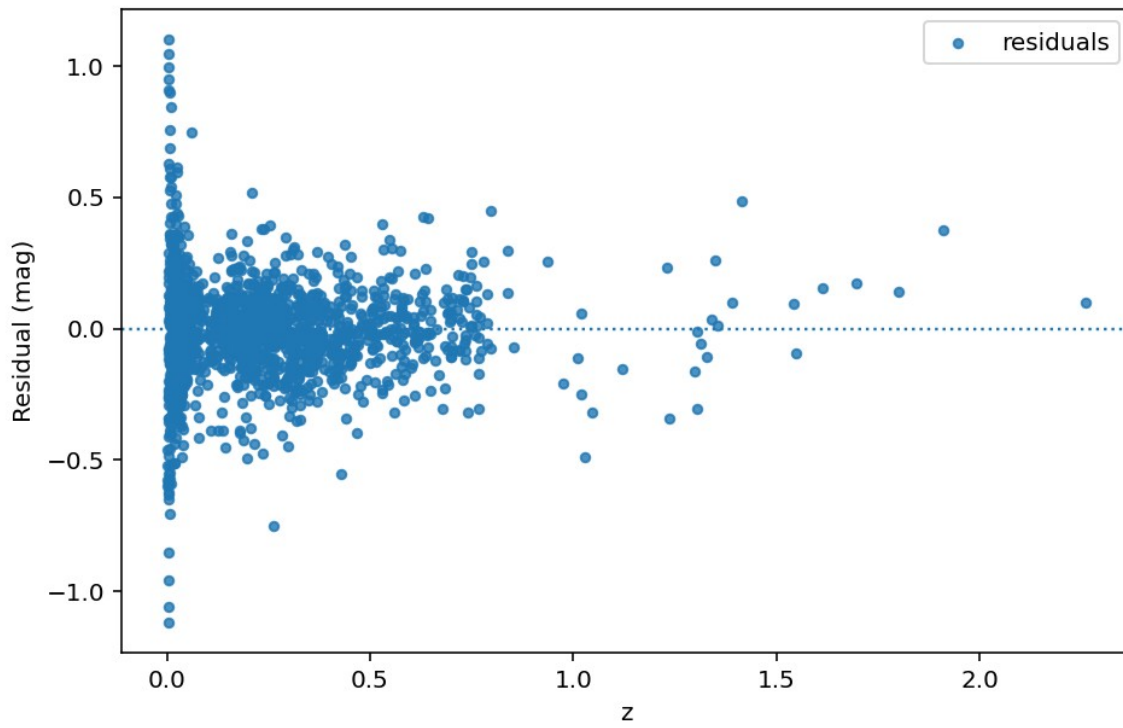
Hubble & residuals (thumbnails)

Λ CDM

LCDM | $H_0=71.87$, $\Omega_m=0.216$, $\Omega_\Lambda=0.373$, $A_{acc}=0.000$, $n_{acc}=0.000$, $\gamma_c=0.000$, $\epsilon_M=0.000$
 $\chi^2/dof=1.166$ ($k=3$, $AIC=1986.2$, $BIC=2002.5$) | frozen: $A_{acc}=0$, $n_{acc}=0$, $\gamma_c=0$, $\epsilon_M=0$

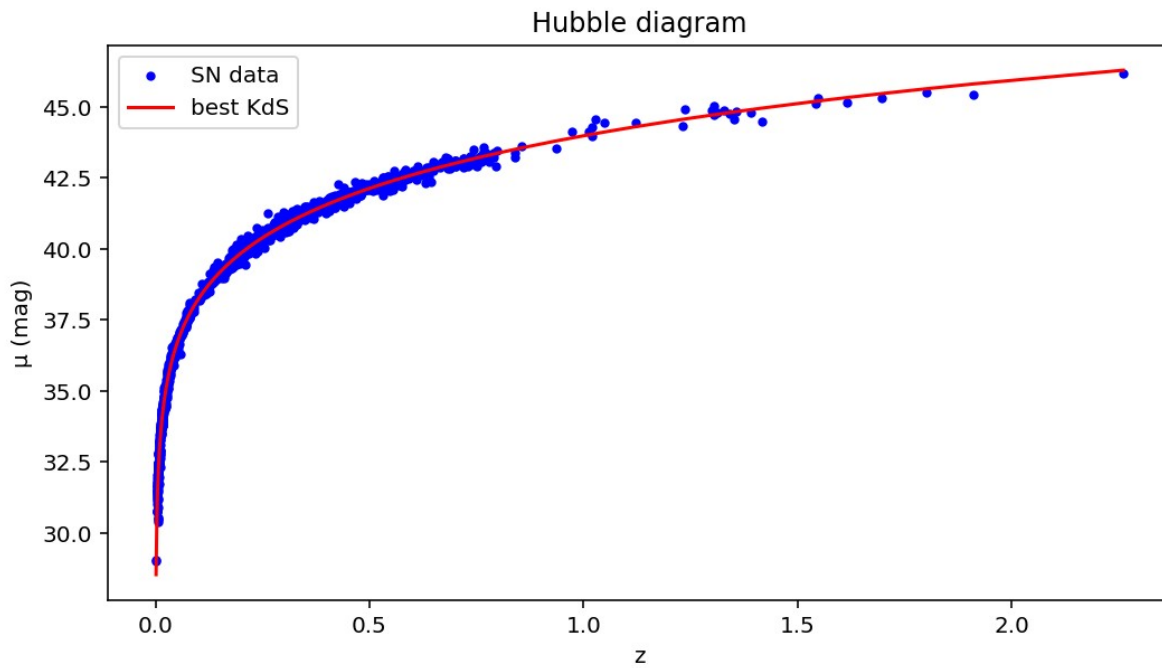


LCDM | $H_0=71.87$, $\Omega_m=0.216$, $\Omega_\Lambda=0.373$, $A_{acc}=0.000$, $n_{acc}=0.000$, $\gamma_c=0.000$, $\epsilon_M=0.000$
 $\chi^2/dof=1.166$ (k=3, AIC=1986.2, BIC=2002.5) | frozen: $A_{acc}=0$, $n_{acc}=0$, $\gamma_c=0$, $\epsilon_M=0$

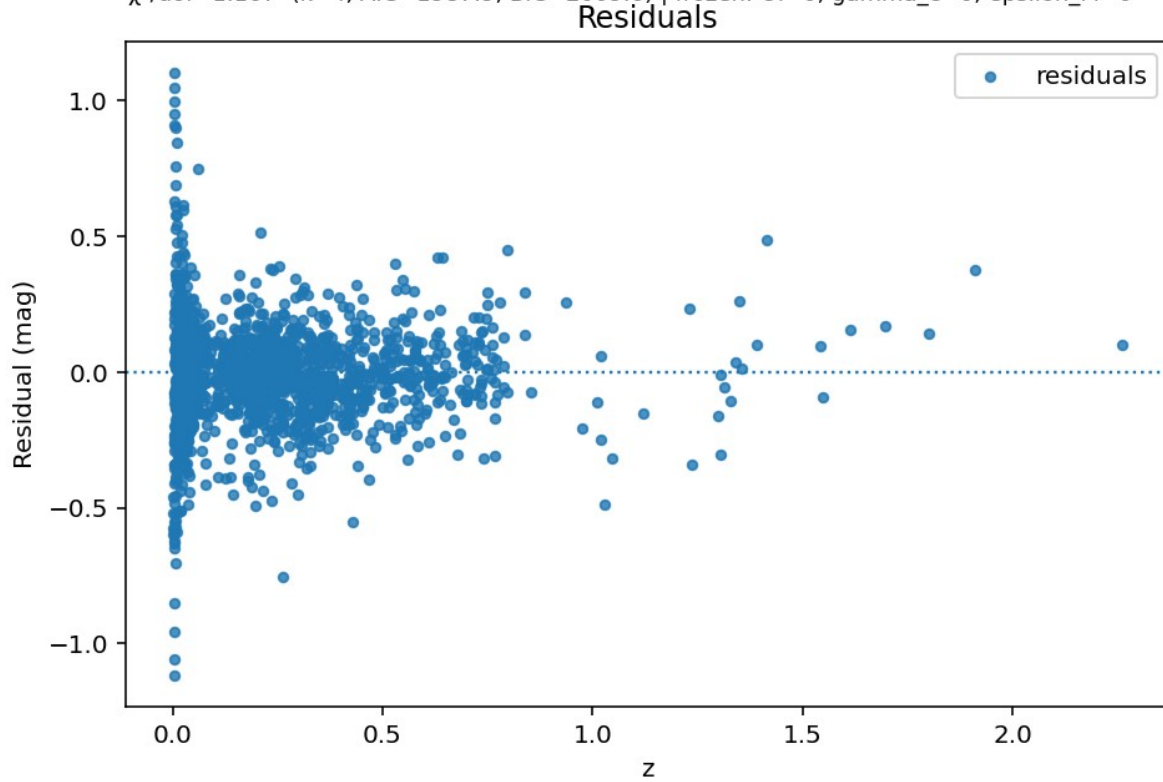


KdS ($\Omega_\Lambda=0$)

KdS | $H_0=50.31$, $\Omega_m=0.396$, $\Omega_\Lambda=0.000$, $A_{acc}=1.038$, $n_{acc}=0.659$, $\gamma_c=0.000$, $\epsilon_M=0.000$
 $\chi^2/dof=1.167$ (k=4, AIC=1987.9, BIC=2009.6) | frozen: $O_l=0$, $\gamma_c=0$, $\epsilon_M=0$

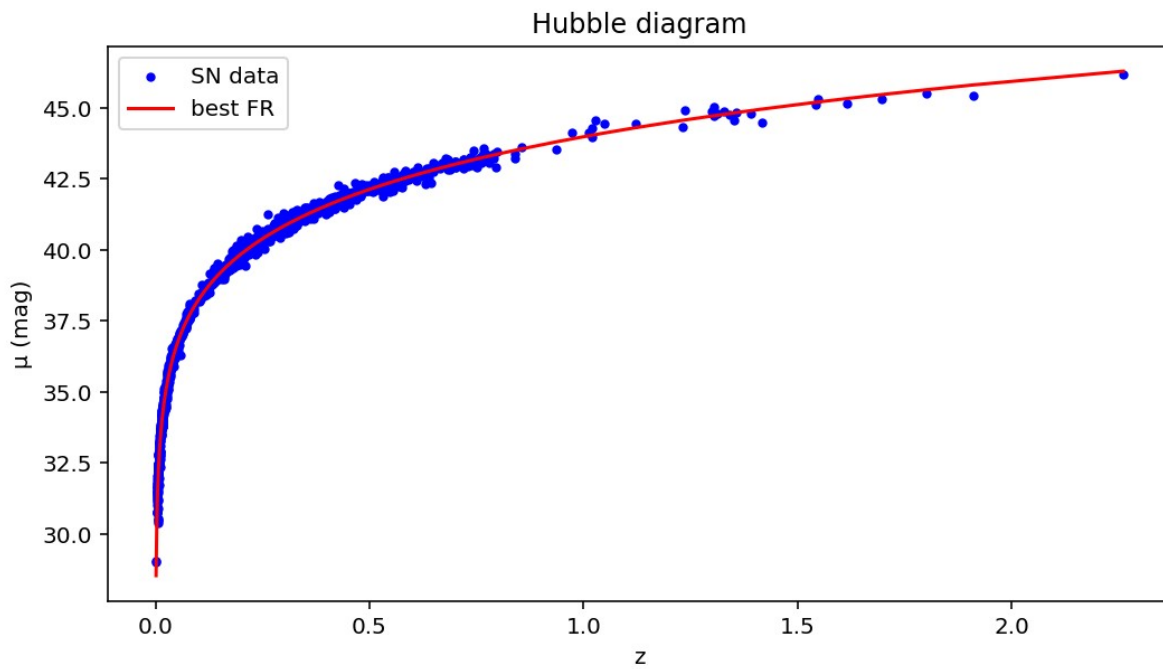


KdS | $H_0=50.31$, $\Omega_m=0.396$, $\Omega_\Lambda=0.000$, $A_{\text{acc}}=1.038$, $n_{\text{acc}}=0.659$, $\gamma_c=0.000$, $\epsilon_M=0.000$
 $\chi^2/\text{dof}=1.167$ ($k=4$, $\text{AIC}=1987.9$, $\text{BIC}=2009.6$) | frozen: $O_l=0$, $\gamma_c=0$, $\epsilon_M=0$

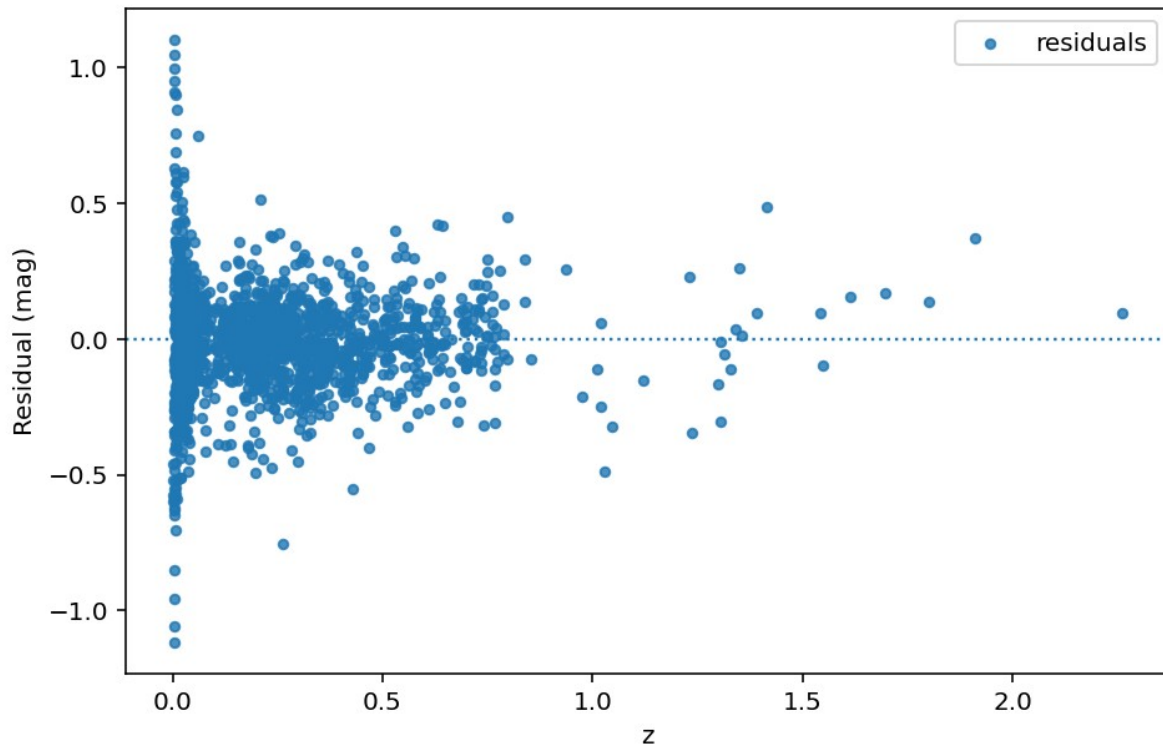


FR (STF-only; $\Omega_\Lambda=0$, $\gamma_c=\epsilon_M=0$)

FR | $H_0=50.40$, $\Omega_m=0.398$, $\Omega_\Lambda=0.000$, $A_{\text{acc}}=1.030$, $n_{\text{acc}}=0.654$, $\gamma_c=0.000$, $\epsilon_M=0.000$
 $\chi^2/\text{dof}=1.167$ ($k=4$, $\text{AIC}=1987.9$, $\text{BIC}=2009.6$) | frozen: $O_l=0$, $\gamma_c=0$, $\epsilon_M=0$

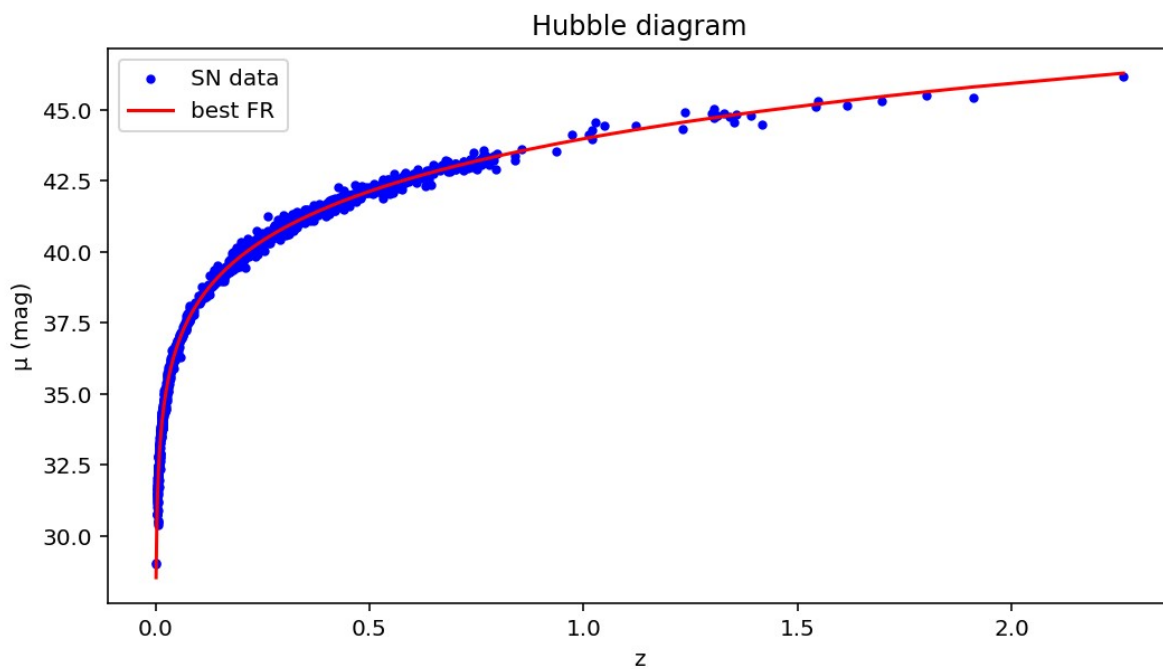


FR | $H_0=50.40$, $\Omega_m=0.398$, $\Omega_\Lambda=0.000$, $A_{\text{acc}}=1.030$, $n_{\text{acc}}=0.654$, $\gamma_c=0.000$, $\epsilon_M=0.000$
 $\chi^2/\text{dof}=1.167$ ($k=4$, $\text{AIC}=1987.9$, $\text{BIC}=2009.6$) | frozen: $O_l=0$, $\gamma_c=0$, $\epsilon_M=0$

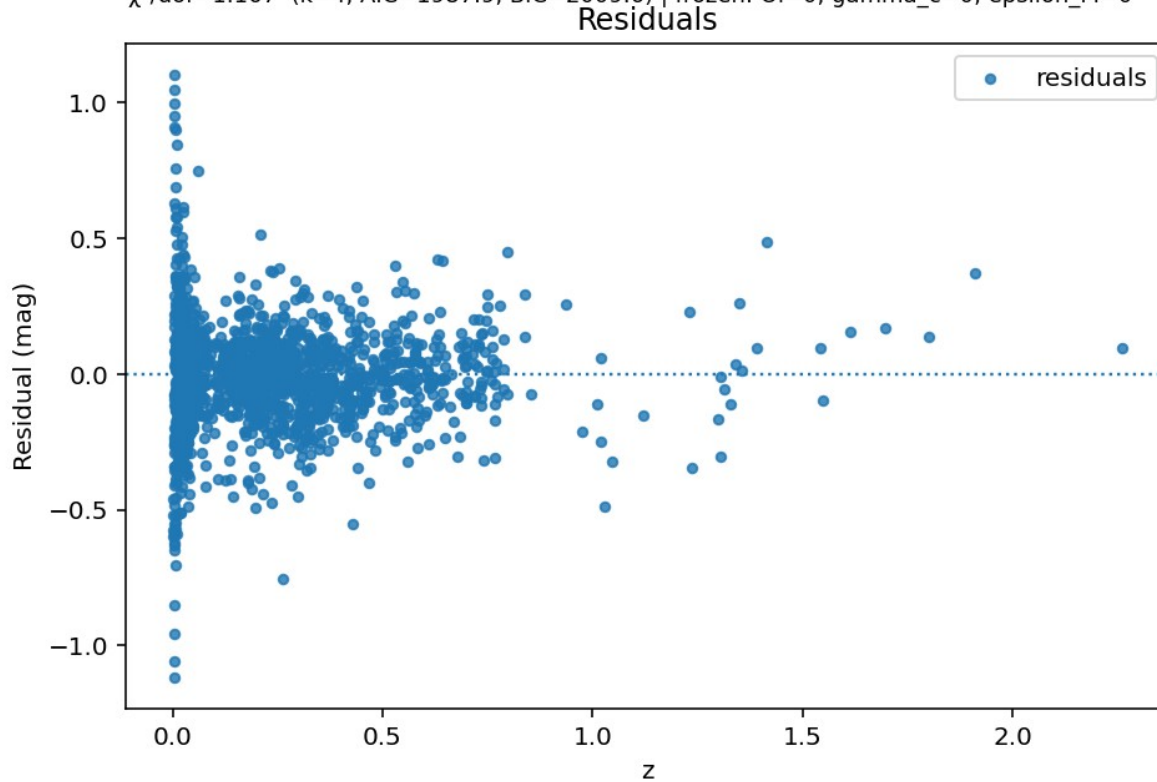


FR (var-c + Λ ; $A_{\text{acc}}=n_{\text{acc}}=\epsilon_M=0$)

FR | $H_0=50.40$, $\Omega_m=0.398$, $\Omega_\Lambda=0.000$, $A_{\text{acc}}=1.030$, $n_{\text{acc}}=0.654$, $\gamma_c=0.000$, $\epsilon_M=0.000$
 $\chi^2/\text{dof}=1.167$ ($k=4$, $\text{AIC}=1987.9$, $\text{BIC}=2009.6$) | frozen: $O_l=0$, $\gamma_c=0$, $\epsilon_M=0$



FR | $H_0=50.40$, $\Omega_m=0.398$, $\Omega_\Lambda=0.000$, $A_{\text{acc}}=1.030$, $n_{\text{acc}}=0.654$, $\gamma_c=0.000$, $\epsilon_M=0.000$
 $\chi^2/\text{dof}=1.167$ ($k=4$, $\text{AIC}=1987.9$, $\text{BIC}=2009.6$) | frozen: $O_l=0$, $\gamma_c=0$, $\epsilon_M=0$



Why residuals fan out at low redshift

Nearby SNe have significant peculiar velocities ($\sigma_v \approx 150\text{--}300$ km/s).

This adds a magnitude error that scales as $1/z$: $\sigma_{\mu}^{\text{PV}}(z) = (5/\ln 10) \cdot \sigma_v / (c \cdot z)$.

Thus dispersion grows rapidly as $z \rightarrow 0$. Coherent local flows and survey heterogeneity add extra width.

The Pantheon(+SH0ES) covariance includes PV and systematics, so nearest SNe are down-weighted accordingly.

Next steps & reference commands

- Isolate variable- c with flatness enforced ($\Omega_k=0$) to test if $\gamma_c \approx 0$ once curvature is removed.
- Isolate SN magnitude evolution (ϵ_M) with both linear- z and $\log_{10} p_z$ forms.
- Optionally add BAO/CC/CMB shift constraints to break SN degeneracies.

Var- c only (no Λ):

```
%run '/Users/boyde/.spyder-py3/PLamb_Test_10V6c.py' \
--model FR \
--data '/Users/boyde/Downloads/Pantheon+SH0ES.dat' \
--cov '/Users/boyde/Downloads/Pantheon+SH0ES_STAT+SYS.cov' \
--Ol 0 --fix_Ol \
```

```
--A_acc 0 --fix_A_acc \
--n_acc 0 --fix_n_acc \
--epsilon_M 0 --fix_epsilon_M \
--sampler emcee --nwalkers 48 --nsteps 5000 --nburn 2000 \
--nint 220 --thin 1 --tight-priors --corner --seed 73 --verbose \
--out 'plamb_runs/10V6c_FR_varC_only_noLambda'
```

Var-M only (no Λ), linear-z:

```
%run '/Users/boyde/.spyder-py3/PLamb_Test_10V6c.py' \
--model FR \
--data '/Users/boyde/Downloads/Pantheon+SH0ES.dat' \
--cov '/Users/boyde/Downloads/Pantheon+SH0ES_STAT+SYS.cov' \
--dM-mode linear_z \
--Ol 0 --fix_Ol \
--A_acc 0 --fix_A_acc \
--n_acc 0 --fix_n_acc \
--gamma_c 0 --fix_gamma_c \
--sampler emcee --nwalkers 48 --nsteps 5000 --nburn 2000 \
--nint 220 --thin 1 --tight-priors --corner --seed 75 --verbose \
--out 'plamb_runs/10V6c_FR_varM_linear_noLambda'
```

Var-M only (no Λ), $\log(1+z)$:

```
%run '/Users/boyde/.spyder-py3/PLamb_Test_10V6c.py' \
--model FR \
--data '/Users/boyde/Downloads/Pantheon+SH0ES.dat' \
--cov '/Users/boyde/Downloads/Pantheon+SH0ES_STAT+SYS.cov' \
--dM-mode log1pz \
--Ol 0 --fix_Ol \
--A_acc 0 --fix_A_acc \
--n_acc 0 --fix_n_acc \
--gamma_c 0 --fix_gamma_c \
--sampler emcee --nwalkers 48 --nsteps 5000 --nburn 2000 \
--nint 220 --thin 1 --tight-priors --corner --seed 76 --verbose \
--out 'plamb_runs/10V6c_FR_varM_log1pz_noLambda'
```